

Detecting Malicious Blockchain Activity

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Data science Project

Background

Crypto

- Blockchain technology caused a financial revolution
 - Instant transfer
 - Unregulated
 - Anonymous

Problem

- Today the technology is used in part for:
 - Money laundering
 - Terror funding
 - Tax evasion

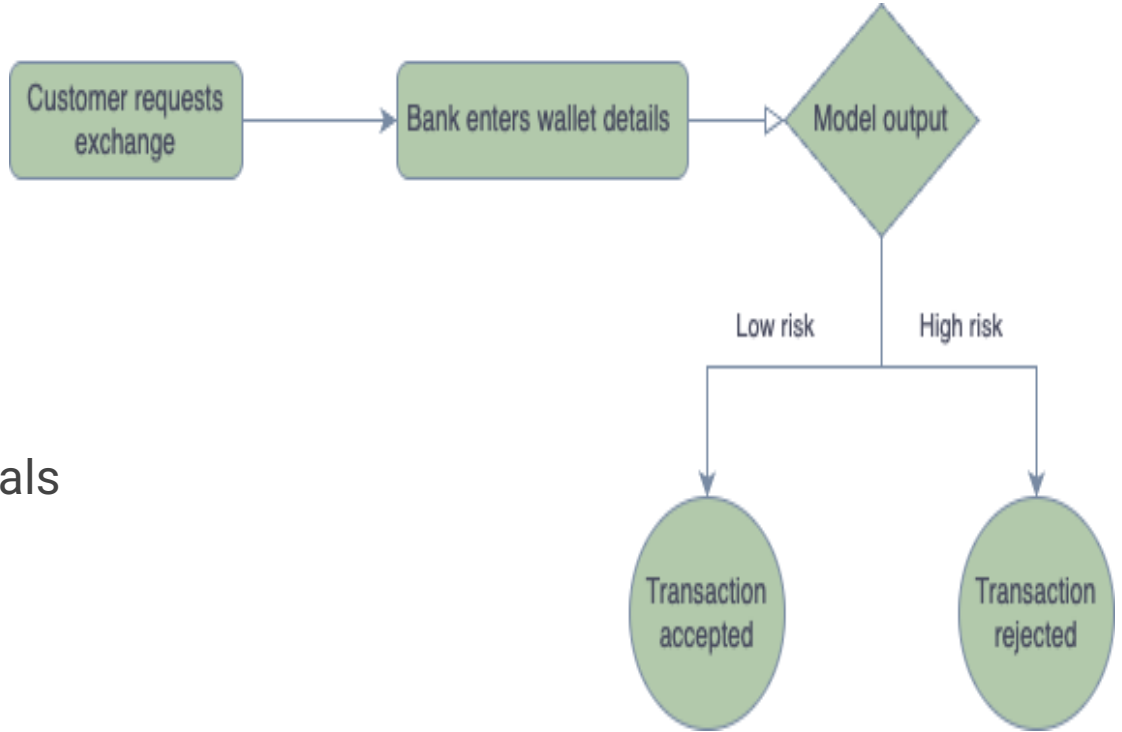
Solution

- There is a need for **automatic tools** to track these behaviours

Our Clients

Our product is made for:

- Financial institutions
- Military and police
- Businesses and individuals





Goals & Challenges

Goals

Build a system that:

- **Interaction** with user
- **Processes** real-time transaction data
- **Classification** of requested wallet
- **Results Showcase** to user



System Architecture

Technology / Architecture	Advantages	Disadvantages / Limitations	Maturity vs. Alternatives	Why this was chosen for the project
Graph Convolutional Network (GCN)	Simple, well-understood; exploits local graph structure; strong baseline.	Does not scale well; oversmoothing; may underperform strong tabular models.	Very mature but less expressive than attention- or dynamic-GNNs.	Canonical baseline enabling comparison with prior AML studies.
GraphSAGE	Inductive; scalable sampling; handles new nodes at inference.	Requires tuning of sampling sizes; may lose global info.	Widely used in industry; mature for large-scale graphs.	Provides scalability + inductive capability for large transaction networks.
Topology-Adaptive GCN layers (TAGCN etc.)	Captures multi-hop information; flexible filters.	More parameters; higher risk of overfitting.	Moderately mature; common in temporal/dynamic GNNs.	Enables richer multi-hop propagation without deep networks.
Attention / Residual GNN variants (GAT / GAT-ResNet ideas)	Learns neighbor importance; residuals stabilize training.	Higher computational cost; more hyperparameters.	Common in graph learning; less mature in AML but strong results.	Adds robustness via attention-like weighting + residuals.
Class-imbalance techniques	Improves recall for rare illicit classes.	Risk of overfitting minorities; graph-aware sampling is hard.	Standard in tabular fraud; emerging in GNNs.	Essential for rare-event AML detection and outperforming baselines.

Literature Review

Paper	Dataset	Core Method	Imbalance Handling	Temporal Modeling	Key Innovation	Performance (F1/AUC on Elliptic or equiv.)
Weber et al. (2019) arxiv	Elliptic (Bitcoin)	Static GCN, EvolveGCN vs. RF/MLP	Class-weighted loss	Basic (time slices in EvolveGCN)	Elliptic dataset release + GCN baselines	RF: 0.93 F1; GCN: 0.90 F1 (illicit class)
Asiri et al. (2025) nature	Elliptic (Bitcoin)	Optimized GCN with feature engineering	Cost-sensitive learning	No	Graph-based evaluation + tuned GCN layers	98.5% accuracy, AUC 0.9444
Zhang et al. (2024) arxiv+1	Elliptic (Bitcoin)	GAT, GAT-ResNet (attention + residuals)	Implicit via attention weighting	No	GAT-ResNet for missing data + global patterns	0.956 precision, MCC 0.733
Alarab & Prakoonwit (2022) d-nb	Elliptic (Bitcoin)	LSTM + TAGCN + MC-dropout active learning	Active learning prioritizes minorities	Yes (LSTM temporal encoding)	Bayesian uncertainty for label-efficient AML	0.95 F1 (supervised); 0.92 w/ 20% labels



Exploratory Data Analysis

Data

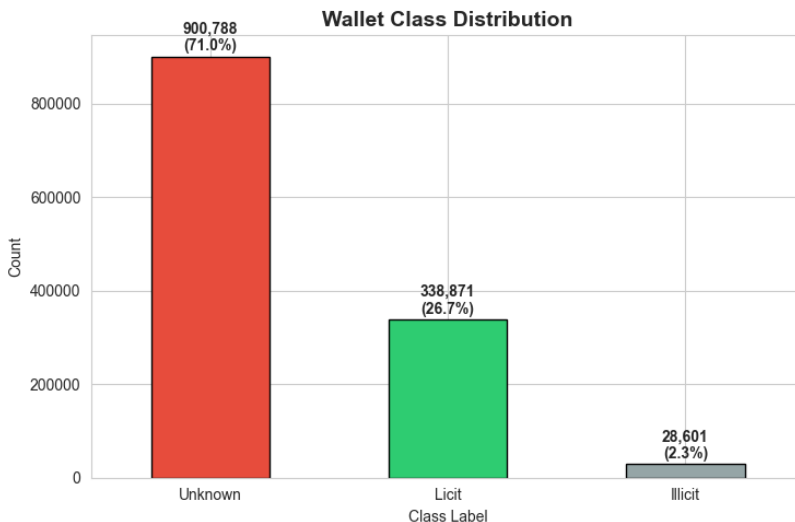
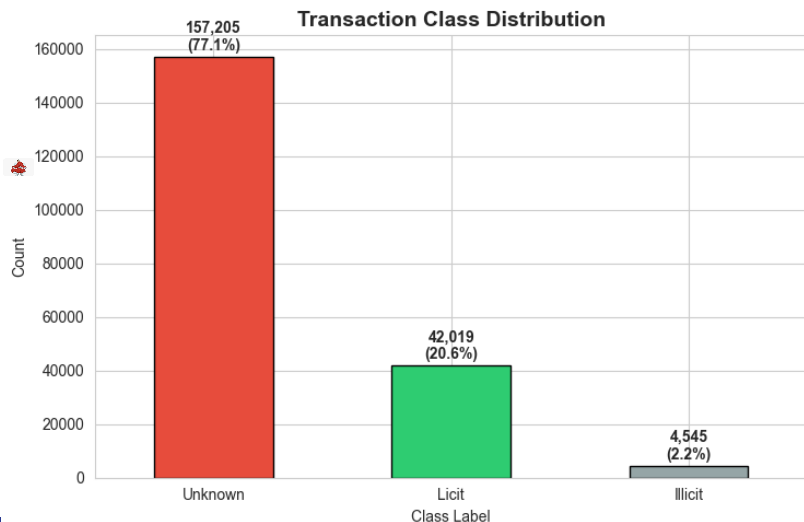
- Elliptic
 - Bitcoin, 203K labeled transactions
 - Anonymized features
- Real-CATS
 - Over 200K labeled wallets
 - Bitcoin & Ethereum
 - 25 Features



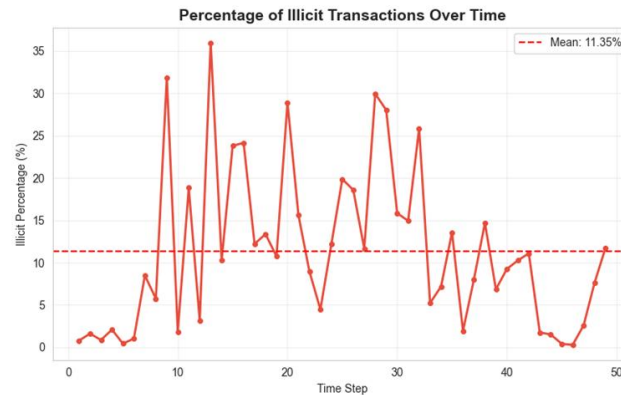
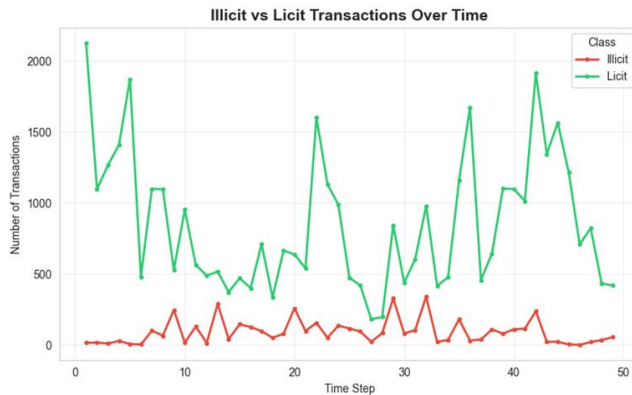
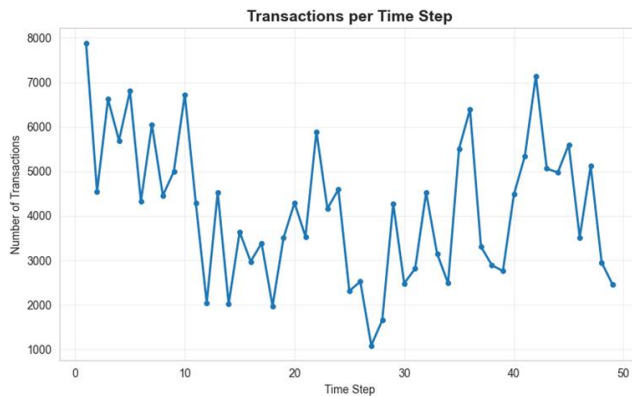
Elliptic ++ - EDA

- Dataset Shapes:

- Transactions: (203769, 186)
- Wallets: (1268260, 59)
- Transaction edges: (234355, 2)

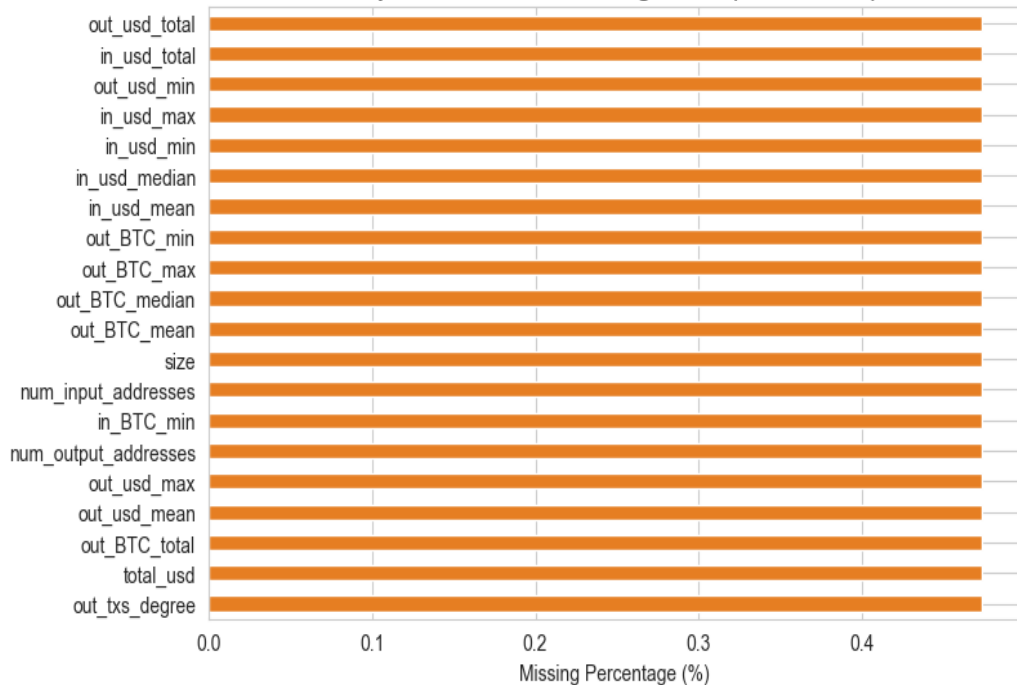


Transactions over time



Missing values

Top 20 Features with Missing Values (Transactions)



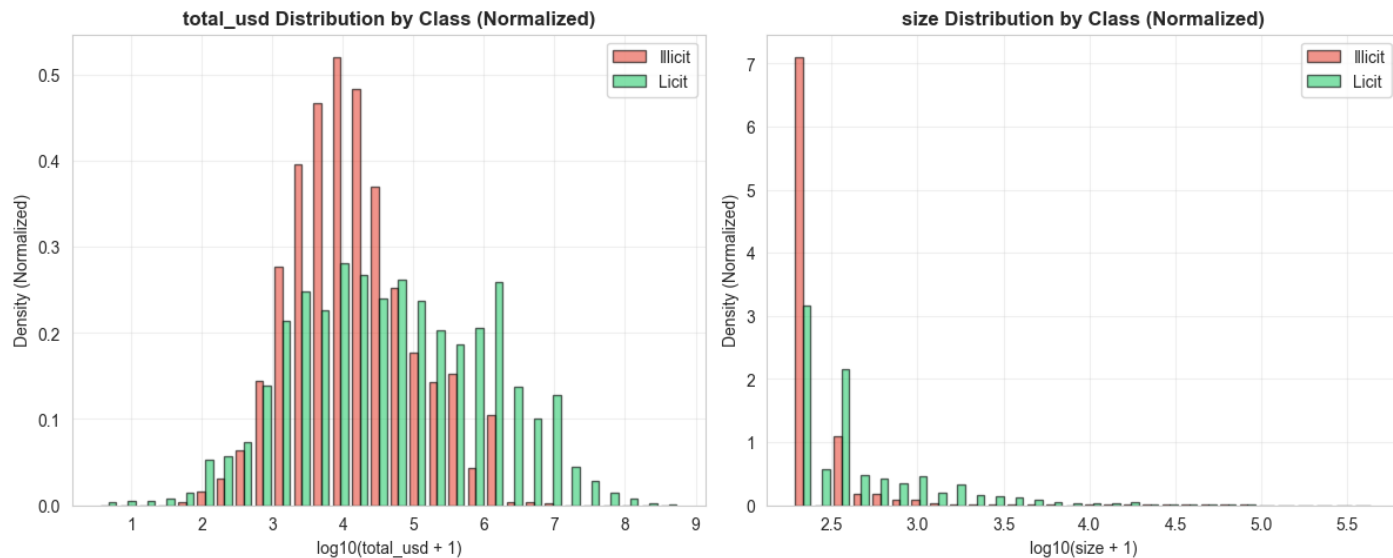
Missing Values (Wallets)





Licit vs Illicit

Feature Distributions Normalized by Class Size - Shows Trends Independent of Class Imbalance



Licit vs Illicit

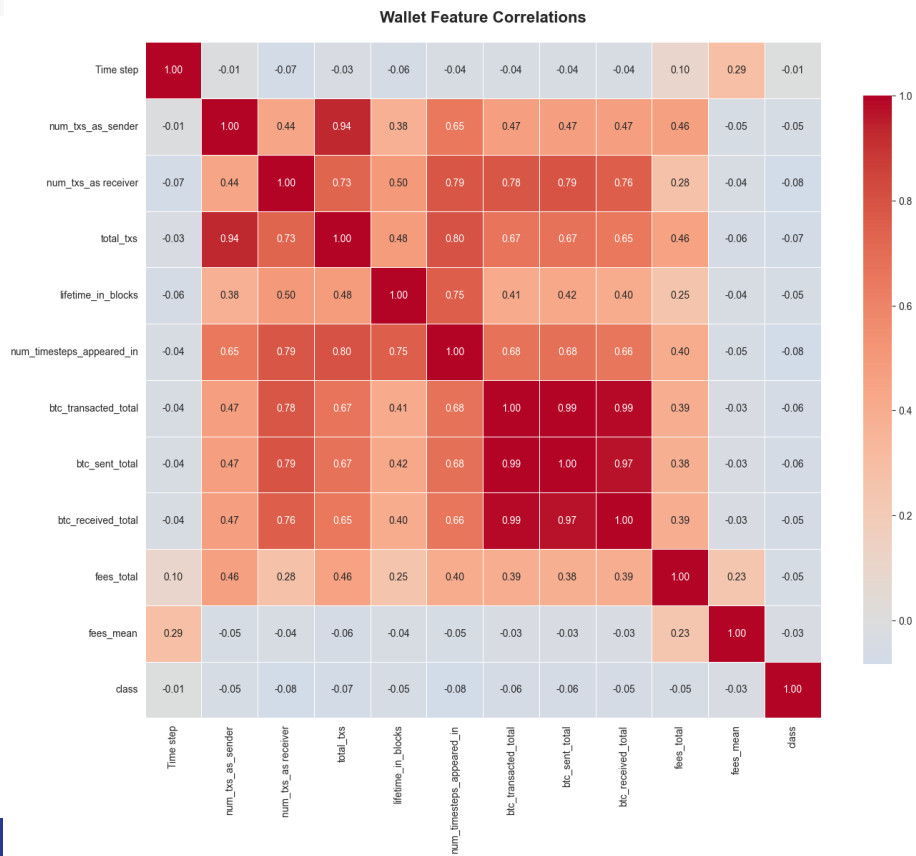
Network Degree Analysis by Class (FULL DATASET)



Class Distribution In Network

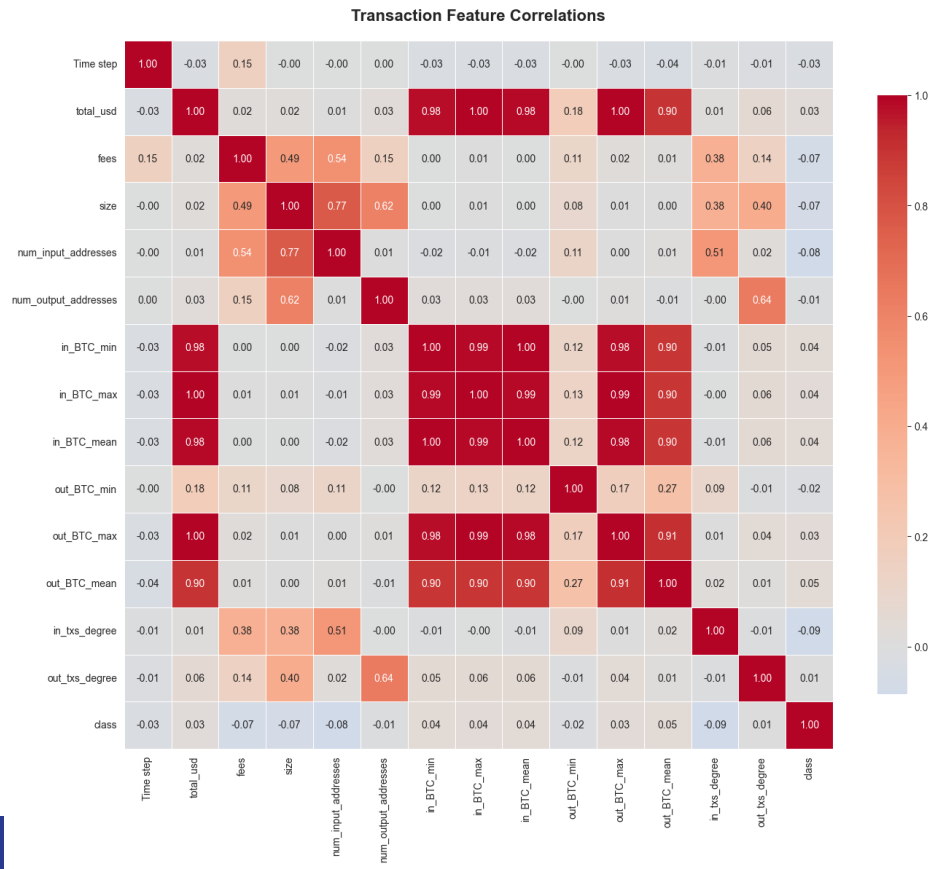
From	To	Count
Unknown	Unknown	131,778
Unknown	Licit	45,386
Licit	Licit	33,930
Licit	Unknown	15,116
Unknown	Illicit	3,993
Illicit	Unknown	1,458
Illicit	Illicit	998
Illicit	Licit	915
Licit	Illicit	781

Elliptic ++ - EDA - Wallet Feature Correlation

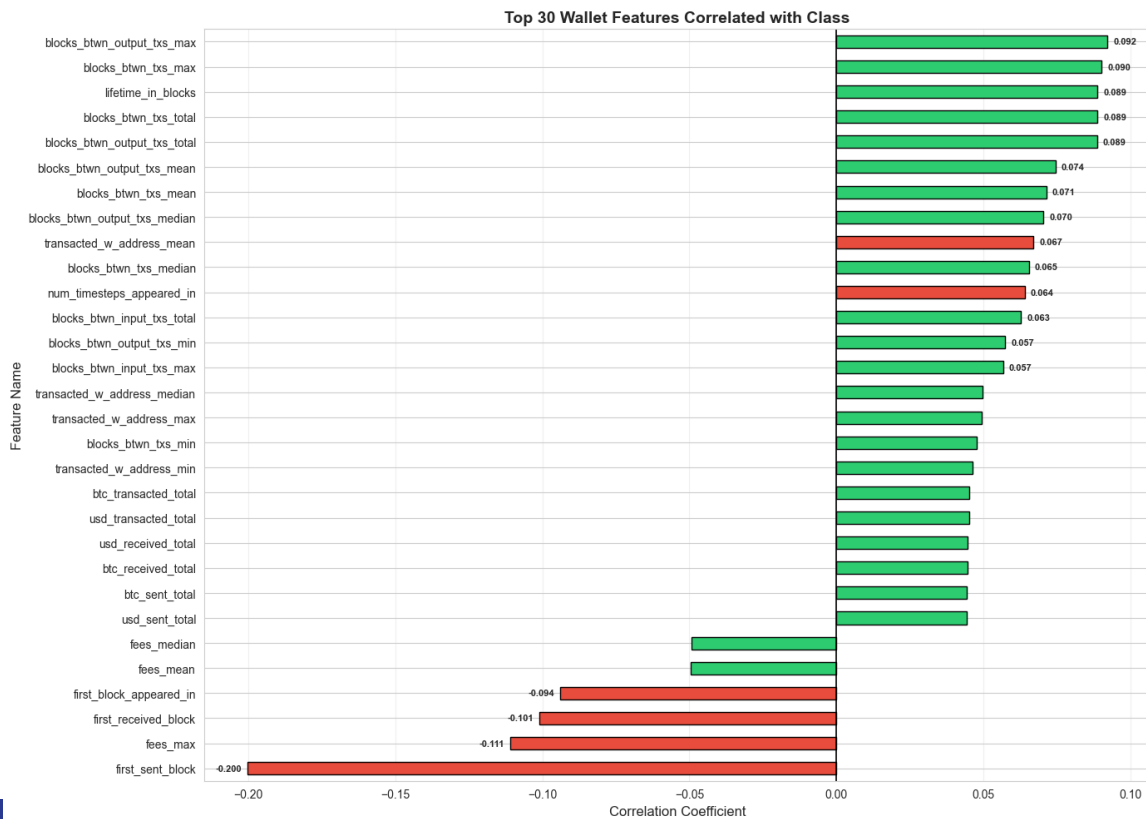




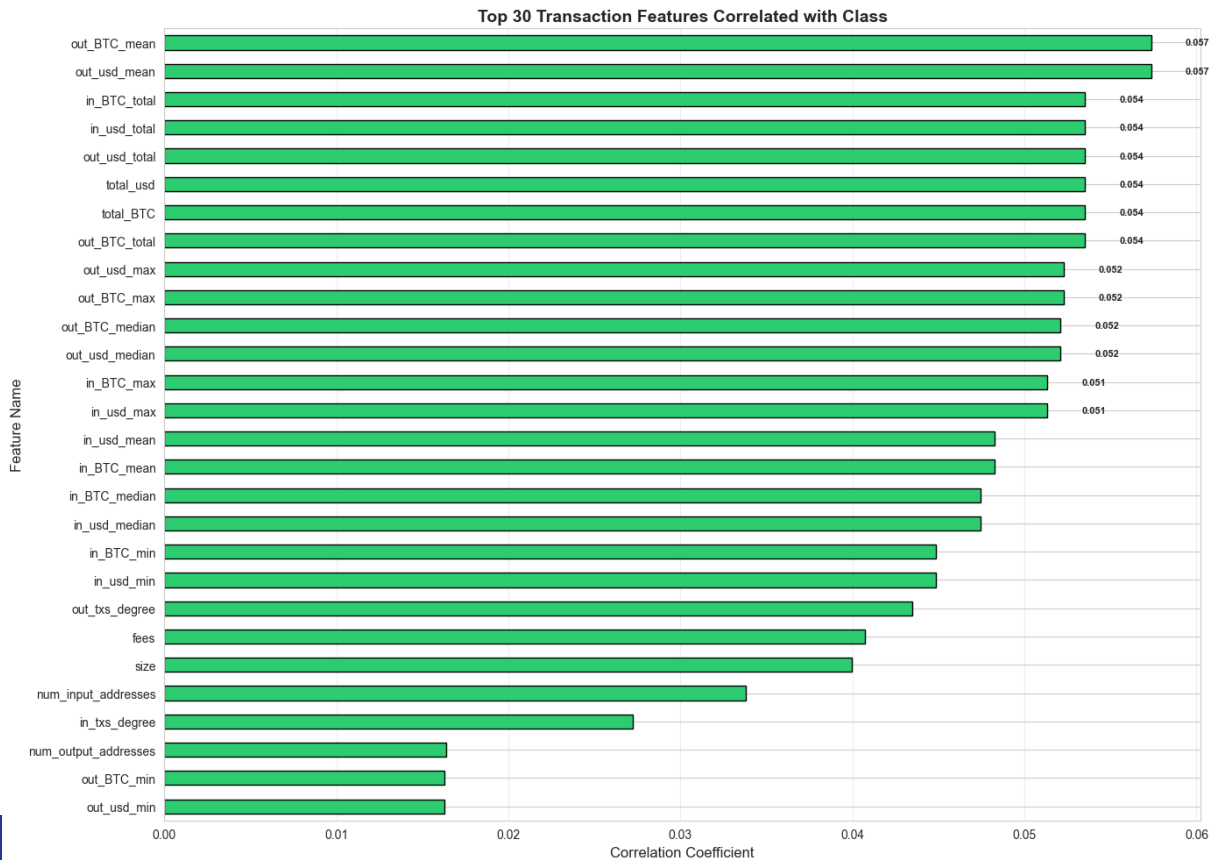
Elliptic ++ - EDA - Transaction Feature Correlation



Elliptic ++ - EDA - Feature Correlation To Class

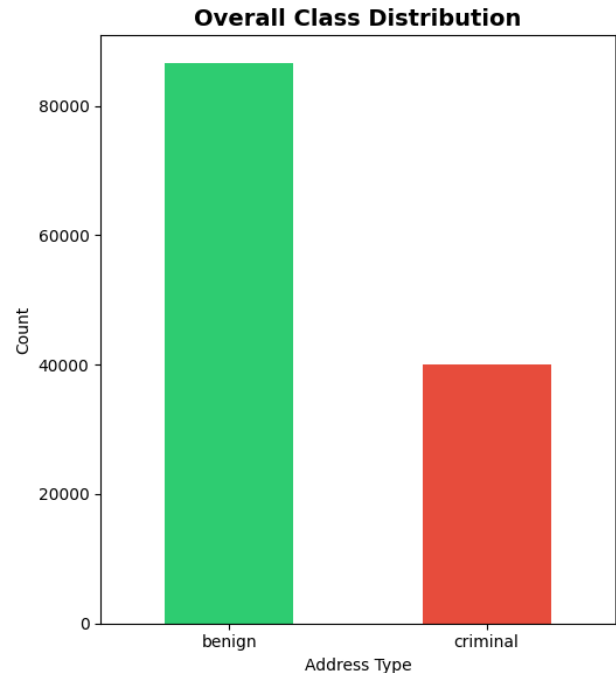


Elliptic ++ - EDA - Feature Correlation To Class



REAL-CATS EDA

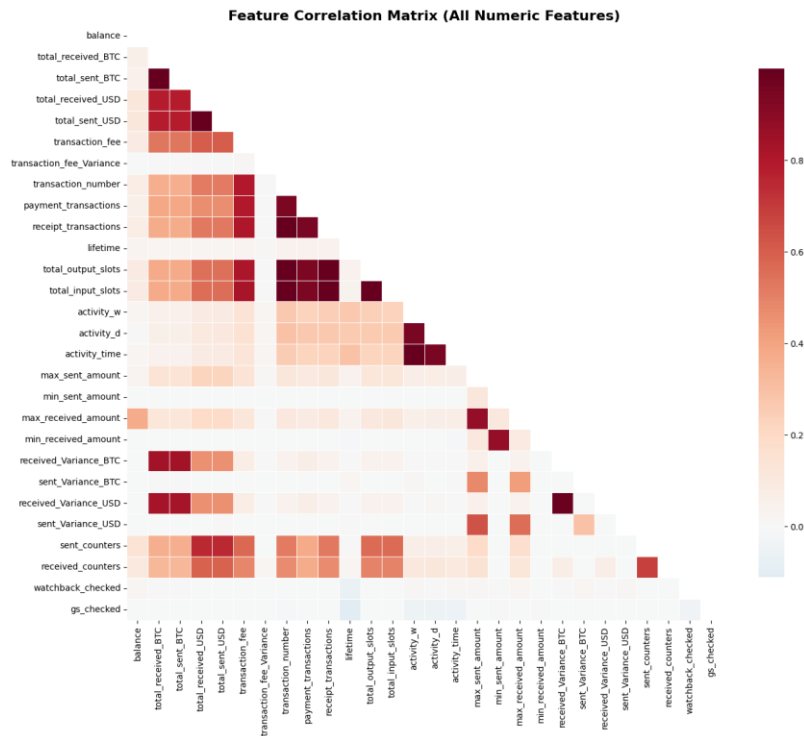
- Data composition:
 - Total addresses: 126,615 Bitcoin addresses
 - Criminal Addresses: 40,032 (31.6%)
 - Benign Addresses: 86,583 (68.4%)
 - Class Balance: 1:2.16 ratio
- Filtering: Removed zero-transaction addresses
- Data quality:
 - Complete Features: 26 out of 37 features
 - High Missing (>10%): 10 features (mostly transaction detail fields)



Transaction Patterns

Metric	Criminal Median	Benign Median	Difference	Significance
Transaction Count	4.0	2.0	2.0x higher	Criminals are more active
Total Received (USD)	\$2,433	\$1,591	53% higher	Criminals have Higher volumes
Total Sent (USD)	\$2,426	\$1,049	131% higher	Criminals send much more money

Feature Correlation



Top Problematic Correlations

1. **totalreceivedBTC ↔ totalsentBTC: 1.000**
2. **totalreceivedUSD ↔ totalsentUSD: 1.000**
3. **totaloutputslots ↔ totalinputslots: 0.999**
4. **transactionnumber ↔ receipttransactions: 0.995**
5. **activityw ↔ activitytime: 0.995**

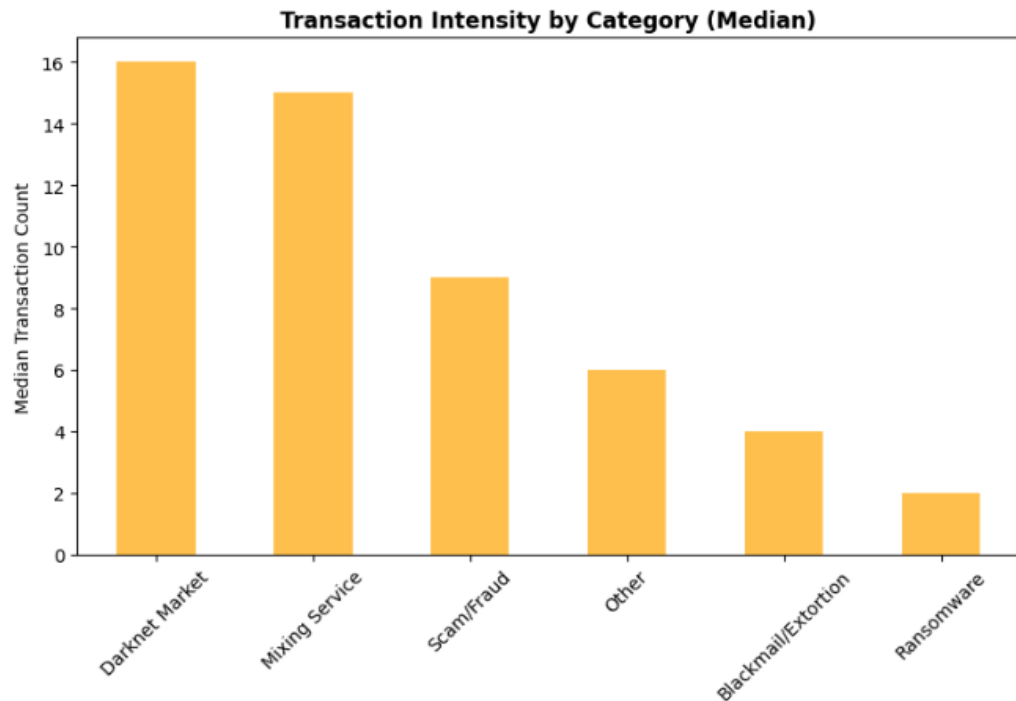
Criminal Address Analysis



Prevalence vs. Perception

- Simple scams and ransomware remain the dominant forms of Bitcoin criminal activity.
- Volume: These "low-tech" crimes represent the vast majority of illicit volume.
- Specialization: Mixing services represent a much smaller, highly specialized segment of the criminal ecosystem.

Criminal Address Analysis



The Metric:

Transaction frequency proxies for organizational sophistication.

Structured:

High frequency; requires continuous liquidity.

Opportunistic:

Low frequency; relies on disposable wallets.